A HORIZONTAL-LINE EMBEDDING ANSWER FOR WILDRICK'S HEISENBERG BOCHNER-DERIVATIVE QUESTIONS

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Status

This packet gives affirmative answers to Problems 5.3, 5.4, and 5.5 in Wildrick [2]. The key point is that the sub-Riemannian Heisenberg group has a 1-Lipschitz retraction onto any chosen horizontal line. Combining that retraction with a Kuratowski embedding gives a Banach-space embedding that is exactly linear on the chosen line.

Source questions

The relevant source passage is on PDF page 5. Problem 5.3 asks whether, for the Heisenberg group, there is any nonconstant map into $\mathbb{H}$ and any bi-Lipschitz embedding into a Banach space for which the Bochner partial derivatives exist. Problem 5.4 asks whether some embedding has a nonzero velocity at some point. Problem 5.5 asks the simpler concrete version: whether there is a bi-Lipschitz embedding $\iota : \mathbb{H} \to V$ and a vector $\vec{v} \in V$ such that

$$\iota(x + 0i, 0) = x\vec{v} \qquad (x \in \mathbb{R}).$$

Date: 2026-07-01.

case of $\mathbb{H}$ is still unknown:

Problem 5.3. Is there *any* non-constant mapping $f\colon \Omega \to \mathbb{H}$ and any bi-Lipschitz embedding $\iota\colon \mathbb{H} \to V$ into a Banach space so that the Bochner partial deriviatives of f with respect to ι exist?

We now consider a closely related issue. Consider a Lipschitz path $c\colon [-1,1] \to \mathbb{H}$. For each point $t \in [-1,1]$ where $\pi \circ c$ is differentiable, there exists (suitably defined) element $c'(t)$ of the measurable tangent space to $\mathbb{H}$ at $c(t)$, called a *velocity* (see [6]). Denote by $\mathcal{V}_p$ the set of velocities in the measurable tangent space to $\mathbb{H}$ a point $p \in \mathbb{H}$, and consider a bi-Lipschitz embedding $\iota\colon \mathbb{H} \to V$. Consider the subset

$$\mathcal{V}_{\iota,p} = \{v \in \mathcal{V}_p : v = c'(0) \text{ and } \iota \circ c \text{ is Fréchet differentiable at } 0\}$$

The methods of [6] and the discussion in Section 2 show that $\dim \mathcal{V}_{\iota,p} \leq 1$ for almost every $p \in \mathbb{H}$. However, it is unclear if $\mathcal{V}_{\iota,p}$ can ever be non-trivial:

Problem 5.4. Is there a bi-Lipschitz embedding $\iota\colon \mathbb{H} \to V$ into a Banach space so that $\mathcal{V}_{\iota,p}$ contains a non-zero velocity for some $p \in \mathbb{H}$?

A simpler version of Problem 5.4, which can be posed without referring to the tools developed in [6], is as follows:

Problem 5.5. Is there a bi-Lipschitz embedding $\iota\colon \mathbb{H} \to V$ of the Heisenberg group into a Banach space V so that there is a single vector $\vec{v} \in V$ satisfying

$$\iota(x + 0i, 0) = x \cdot \vec{v}$$

for all $x \in \mathbb{R}$?

Proof Intuition

A Kuratowski embedding remembers all distances, but it has no reason to be linear on a selected geodesic. If the selected geodesic is the range of a 1-Lipschitz retraction r, we can keep the linear coordinate on the geodesic and use the difference

$$\kappa(x) - \kappa(r(x))$$

for the remaining coordinates. This difference vanishes on the geodesic. It still controls the lost metric information because

$$\kappa(x) - \kappa(y) = \big(\kappa(x) - \kappa(r(x))\big) - \big(\kappa(y) - \kappa(r(y))\big) + \big(\kappa(r(x)) - \kappa(r(y))\big).$$

Thus the original distance is bounded by the new transverse part plus the linear retraction coordinate.

A retract lemma

Lemma 1. *Let (X, d) be a separable metric space. Suppose that $\ell : \mathbb{R} \to X$ is an isometric embedding and that $r : X \to \ell(\mathbb{R})$ is a 1-Lipschitz retraction. Write $a : X \to \mathbb{R}$ for the coordinate defined by $r(x) = \ell(a(x))$. Then there is a Banach space V and a bi-Lipschitz embedding $I : X \to V$ such that*

$$I(\ell(s)) = sv \qquad (s \in \mathbb{R})$$

for a nonzero vector $v \in V$.

Proof. Let $\kappa : X \to \ell^\infty$ be any Kuratowski isometric embedding. Set

$$V = \mathbb{R} \oplus_\infty \ell^\infty$$

and define

$$I(x) = \big(a(x),\, \kappa(x) - \kappa(r(x))\big).$$

If $x = \ell(s)$, then $r(x) = x$ and $a(x) = s$, so $I(\ell(s)) = (s, 0) = s(1, 0)$. Put $v = (1, 0)$.

Since r is 1-Lipschitz and ℓ is isometric, the coordinate a is 1-Lipschitz. The second component is 2-Lipschitz, because

$$\|\kappa(x) - \kappa(r(x)) - \kappa(y) + \kappa(r(y))\|_\infty \le d(x, y) + d(r(x), r(y)) \le 2d(x, y).$$

Hence I is 2-Lipschitz.

For the lower estimate, use that κ is isometric:

$$\begin{aligned} d(x, y) &= \|\kappa(x) - \kappa(y)\|_\infty \\ &\le \|\kappa(x) - \kappa(r(x)) - \kappa(y) + \kappa(r(y))\|_\infty + \|\kappa(r(x)) - \kappa(r(y))\|_\infty \\ &= \|\kappa(x) - \kappa(r(x)) - \kappa(y) + \kappa(r(y))\|_\infty + |a(x) - a(y)| \\ &\le 2\|I(x) - I(y)\|_V. \end{aligned}$$

Thus $\|I(x) - I(y)\|_V \ge d(x, y)/2$, and I is bi-Lipschitz. □

Application to the Heisenberg group

Let $\mathbb{H}$ be the first Heisenberg group with its standard CC metric d_{cc}. Let

$$\ell(s) = (s + 0i, 0).$$

This is an isometric copy of $\mathbb{R}$: the horizontal curve $s \mapsto (s+0i, 0)$ has length equal to the Euclidean length, and every horizontal curve joining $\ell(s)$ to $\ell(t)$ has length at least $|s - t|$ because its real horizontal coordinate changes by $s - t$.

Define

$$r(z, t) = (\operatorname{Re} z + 0i, 0).$$

The same length estimate shows that r is 1-Lipschitz, since

$$d_{cc}(r(z, t), r(w, u)) = |\operatorname{Re} z - \operatorname{Re} w| \le d_{cc}((z, t), (w, u)).$$

Therefore Lemma 1 applies to $\mathbb{H}$.

Theorem 1. *There exists a Banach space V, a bi-Lipschitz embedding $I : \mathbb{H} \to V$, and a nonzero vector $v \in V$ such that*

$$I(x + 0i, 0) = xv \qquad (x \in \mathbb{R}).$$

Thus Problem 5.5 of [2] *has an affirmative answer.*

Proof. Apply Lemma 1 to the horizontal line ℓ and the retraction r above. □

Consequences for Problems 5.3 and 5.4

Corollary 1. *Problem 5.4 of* [2] *has an affirmative answer.*

Proof. Let I be the embedding in Theorem 1. For the horizontal path $c(t) = (t + 0i, 0)$, one has $I(c(t)) = tv$, so $I \circ c$ is Frechet differentiable at 0 with derivative $v \ne 0$. In the usual Cheeger differentiable structure on $\mathbb{H}$, this path has the nonzero horizontal velocity in the real direction at the identity. Hence $\mathcal{V}_{I,0}$ contains a nonzero velocity. □

Corollary 2. *Problem 5.3 of* [2] *has an affirmative answer. In fact, for every bounded Euclidean domain* $\Omega \subset \mathbb{R}^n$ *there is a nonconstant Lipschitz map* $f : \Omega \to \mathbb{H}$ *and a bi-Lipschitz embedding* $I : \mathbb{H} \to V$ *such that the Bochner partial derivatives of* $I \circ f$ *exist and are integrable.*

Proof. Use the embedding I from Theorem 1 and define

$$f(x_1, \ldots, x_n) = (x_1 + 0i, 0).$$

Then f is Lipschitz and nonconstant on any Euclidean domain. Moreover

$$I(f(x_1, \ldots, x_n)) = x_1 v.$$

Thus $I \circ f$ has the constant Bochner partial derivative v in the first coordinate and zero Bochner partial derivatives in all other coordinates. Since Ω is bounded, these derivatives are integrable. □

Scope and novelty

This result answers only the Heisenberg existence questions 5.3, 5.4, and 5.5. It does not characterize the Poincare-inequality spaces in Problem 5.1, and it does not characterize the metric spaces with an ι-Radon-Nikodym embedding in Problem 5.2. It also does not contradict the source's main theorem: the construction gives one good family of maps along one horizontal line, while the main theorem rules out embeddings that make every Lipschitz map into a Cheeger fractal Bochner differentiable.

Before promotion, the cheap run indexes were searched for the arXiv id, the source title, Bochner partial derivatives, Cheeger-Kleiner differentiability, ι-Radon-Nikodym terminology, nonzero velocity, and the exact Problem 5.5 horizontal-line formula. No duplicate packet or prior attempt was found. Local full-source fixed-string searches found only the source questions themselves. A bounded web search on 2026-07-01 for the exact horizontal-line formula and the source title with "single vector" did not surface a later answer.

Human review recommendation

The proof is metric and should be checked mainly at one point: under the standard Carnot-Caratheodory normalization, the coordinate retraction $(z, t) \mapsto (\operatorname{Re} z + 0i, 0)$ is 1-Lipschitz. Once this is accepted, the retract lemma gives the bi-Lipschitz embedding formally.

References

[1] J. Heinonen, *Lectures on analysis on metric spaces*, Universitext, Springer, 2001.
[2] K. Wildrick, *Bochner Partial Derivatives, Cheeger-Kleiner Differentiability, and Non-Embedding*, arXiv:2307.10857, 2023.